Bryan Zang

bryanzang | bryan-zang | bzang19@outlook.com | (647)-393-9656

Skills

Programming

- Python, SQL, R, Matlab, Julia, Java, JavaScript, C, LaTex, PySpark, Numpy, Pandas, sci-kit, TensorFlow

Technologies

- Databricks, Excel, Fabric, PowerBI, Jupyter, Git, Copilot, Refinitiv, Bloomberg Terminal

Work Experience

Research Assistant — *Bank of Canada, Operational Risk and Resilience Analysis* Apr 2026 – Present

- Identified and analyzed key risk indicators to leverage Canadian Overnight Repo Rate Average (CORRA) against market risk.
- Streamlined big data ingestion frameworks to ensure risk compliance.
- Developed agentic tools to analyze risk framework documentation and compliance implementations.
- Conducted and validated structural tests on swap operations between various currencies, including Canadian Dollar (CAD) and Great British Pound (GBP).

Research Assistant — *Bank of Canada, Debt Management* Jul 2025 – Present

- Analyzed financial datasets to obtain insights used in **monetary and fiscal policy** decisions and reports, such as the Debt Management Report (DMR) and Financial Stability Report (FSR).
- Developed tools and metrics for systematic market functionality analysis, such as daily weighted specialness of bonds to indicate demand and scarcity for a security of interest based on trade volume.
- Optimized runtime efficiency of data pipelines by **100 fold** using code parallelization and data structure serialization.
- Quantified **hedge fund positions** and **foreign central banks' holdings** of Government of Canada (GoC) securities using comprehensively derived time series data.
- Created reports on GoC bond market fluctuations to communicate with the Canadian Department of Finance.

Education

University of Waterloo — Bachelor of Mathematics Jun 2024

Majors in Honours Statistics and Honours Computational Mathematics
Minor in Computing

- **Relevant courses:** Probability, Statistics, Neural Networks, Microeconomics, Macroeconomics, Advanced Regression, Estimation & Hypothesis Testing, Stochastic Processes, Databases, Algorithms, Linear/Nonlinear Optimization, Generalized Linear Models

Projects

Spatial Models for Canadian Temperature Inference

- Compared **3 different spatial models** to explain annual Canadian temperature for statistical inference.
- Built models using **spline-based regression**, **polynomial regression**, and **General Additive Models (GAMs)**.
- Measured results using Generalized Cross Validation score (GCV) and R-squared where accuracy for GAMs was highest overall **scoring 17.83 and 0.89** respectively.

Simplex Implementation

- Implemented various theories and computations from *A Gentle Introduction to Optimization* (Guenin, Konemann, and Tuncel).
- Applied theories including **Simplex algorithm**, **Bland's Rule**, **two-phase Simplex**, and **duality theory**.

Tumor Modeling with HTCs and iNKT Cells

- Extended and analyzed a differential equations (DE) system describing interactions between tumor, helper T, and CD8+ cells.
- Conducted **phase plane analysis**, **Hopf Bifurcation**, and **apoptosis analysis**.
- Simulated results for computational analysis based on predator-prey physics.

Stock Trend Predictor

- Used SwiftUI to develop an iOS app that employs several APIs to search and retrieve stock information **under 500ms**.
- Applied computer vision (CV) techniques for **time series trend analysis**.
- Utilized **AWS Lambda** to download relevant stock data, run a prediction model trained using **machine learning (ML)** forecasting techniques, and store outputs in **AWS DynamoDB**.

Certifications

Microsoft Certified: Azure AI Fundamentals — *Microsoft*

- Recognized for fundamental knowledge of machine learning (ML) and artificial intelligence (AI) concepts and familiarity with technologies such as Kubernetes and other related basic cloud services.
- Applied concepts towards natural language processing (NLP), facial recognition, object detection, and image classification.